descriptive_statistics

```
# Module descriptive statistics
# Goal : Summarize data with stats package

# Load data
library("nycflights13")
library(dplyr)
# library("stats")

airlines <- nycflights13::airlines
airports <- nycflights13::airports
flights <- nycflights13::flights
planes <- nycflights13::planes
weather <- nycflights13::weather

# Describe a dataset
table(flights$carrier)
summary(flights$distance)

# Get overall mean +/- sd of distance by carrier p
stats <- flights %>%
 dplyr::group_by(carrier) %>%
 dplyr::summarise(n_flights=n(),
           mean_distance=mean(distance,na.rm = T),
           sd_distance=sd(distance,na.rm = T))

# Calculate the coefficient of correlation between arr_delay and distance in flights dataset
cor(flights$arr_delay,flights$distance,method = "spearman",use = "complete.obs")

# Calculate the quartiles : 0.05, 0.25, 0.5 (median), 0.75 and 0.95 of parameter temp in weather dataset
quantile(flights$arr_delay,probs=c(0.25,0.5,0.75),na.rm=T)

# Calculate the number and % of flight by origin and carrier with dplyr in flights dataset
stats <- flights %>%
 group_by(origin,carrier) %>%
 summarise(n_flights=n(),
       pct=n_flights/sum(n_flights))
```

DESCRIPTIVE TABLE

Type de Paramètre	Format de Présentation Requis
Continu (Distribution Normale)	Moyenne ± Écart-type
Continu (Distribution Non-Normale)	Médiane (Quartile 1 ; Quartile 3)
Catégoriel	Effectif (Pourcentage) *Ex: 15 (20%) pour chaque modalité*

```
# Use dplyr package to print in the console the characteristics of the airport line 345 in table airports
airports %>%
 slice(345)

# Use dplyr package to print in the console the wind characteristics (direction, speed and gust) of 2023-12-10 at 10pm at JFK
weather %>%
 filter(origin=="JFK" & month==12 & day==10 & hour==22) %>%
 select(wind_speed,wind_dir,wind_gust)

# Use dplyr package to summarise for each departure airport the number of flights, minimum distance, maximum distance, average dis
# (Hint : use mean(), min(), max(), sd() base functions with a group_by)

flights %>%
 group_by(origin) %>%
 summarise(n_flights=n(),
     min_distance=min(distance,na.rm=T),
     mean_distance=mean(distance,na.rm=T),
     sd_distance=sd(distance,na.rm=T),
     max_distance=max(distance,na.rm=T))

# Use dplyr package to calculate the speed in miles/hour of each flight in a new variable called speed in table flights
flights <- flights %>%
 mutate(speed=60*(distance/(hour*60+minute)))

# What was the average departure delay in each month ?
# Save this as a data frame `dep_delay_by_month`
# Hint: you'll have to perform a grouping operation then summarizing your data
dep_delay_by_month <- flights %>%
 group_by(month) %>%
 summarise(n_flights=n(),
     mean_dep_delay=mean(dep_delay,na.rm=T))

# Which month had the greatest average departure delay?

dep_delay_by_month %>%
 dplyr::arrange(desc(mean_dep_delay)) %>%
 slice(1)
```

Count the number of records of planes and the sum of seats by company (display real name and not the code) and display it in a table

```
```{r dplyr}
# Use DT::datatable
count_records <- flights %>%
 select(tailnum,carrier) %>%
 unique(.) %>%
 merge(planes,by = "tailnum") %>%
 merge(airlines,by = "carrier") %>%
 group_by(name) %>%
 summarise(n_planes=n(),
    sum_seats=sum(seats))

DT::datatable(count_records)
```
```

Summarise model

```
summary(complete_model)
plot(complete_model)
hist(complete_model$residuals)
shapiro.test(complete_model$residuals)
ggpubr::ggqqplot(complete_model$residuals)
anova(complete_model)
```

Prediction

```
predict(final_model,
      newdata = data.frame(
        cyl="4",
        hp=200,
        wt=4
      ),interval = "prediction")
```

Plotly

Plot the relationship between departure delay (dep_delay) and arrival delay (arr_delay) for flights from JFK airport and color points by company and add a regression line for each company

```
```{r plotly, echo=FALSE}
data_corr <- merge(flights,airlines,by="carrier")
plot_scatter <- ggplot(data=data_corr,
 aes(x=dep_delay,y=arr_delay,
 colour = name)) +
 geom_point(alpha=0.5) +
 geom_smooth(method = "lm")+
 labs(title = "Relationship between dep_delay and arr_delay / compagny",
 colour="name")+
 xlab("dep_delay (minutes)")+
 ylab("arr_delay (minutes)")+
 theme_bw()
ggplotly(plot_scatter)
```
```

ggplot

```
# Plot the distribution with histogram (geom_histogram) of price by cut (5 levels)
# Store the plot in an object called plot_histogram_price

ggplot(data=diamonds,
     aes(x=price,fill = cut)) +
 geom_histogram(alpha=0.5) +
 theme_bw()

# Plot the correlation between price and carat (geom_point) of price by cut (5 levels)
# Store the plot in an object called plot_scatter
# Add a regression line by cut (use geom_smooth)
# Change the colors of the cut levels with : darkred, red, orange, green, darkgreen (use scale_colour_manual)
# Add a title, subtitle and put titles of axes in uppercase and change title of the legend
# Use black and white theme (theme_bw)

cols <- c("darkred","red","orange","green","darkgreen")

plot_scatter <- ggplot(data=diamonds,
           aes(x=price,
             y=carat,
             colour = cut)) +
 geom_point(alpha=0.5) +
 geom_smooth(method = "lm")+
 scale_colour_manual(values = cols)+
 labs(title = "Relationship between price in $, carat by quality of diamonds",
    subtitle = "Number of diamonds : 53940",
    colour="Cut")+
 xlab("PRICE")+
 ylab("CARAT")+
 theme_bw()
```

Normal Dist

```
normal <- rnorm(n = 1500,mean = 15,sd = 5)
hist(normal)
```

Binomial Dist

```
binomal <- rbinom(n = 250,size = 100,prob = 0.75)
hist(binomal)
```

Univariate Linear model

```
model_log <- lm(mpg_log ~ disp,
        data=data)
```

Merge

```
top_5_companies <- flights %>%
 merge(airlines,by = "carrier") %>%
 group_by(carrier,name) %>%
 dplyr::summarise(n=n()) %>%
 arrange(desc(n)) %>%
 ungroup() %>%
 slice(c(1:5))
```

Skewness & Kurtosis

```
weather %>%
 filter(origin=="JFK") %>%
 select(humid) %>%
 as.data.frame()

skewness(weather[weather$origin=="JFK","humid"])
kurtosis(weather[weather$origin=="JFK","humid"])

temp_JFK <- weather %>%
 filter(origin=="JFK")

hist(temp_JFK$humid)
```

Test normality of parameter temperature in april and november

```
shapiro.test(temp_april_november[temp_april_november$month==4,"temp"])
qqnorm(temp_april_november[temp_april_november$month==4,"temp"])
```

Multivariate Linear Model

Select the best model with stepAIC function

```
final_model <- MASS::stepAIC(complete_model,
              direction = "both")
```

Tests : SHAPIRO , BARTLETT , KRUSKAL - temperature mean or median

```
weather_august <- weather %>%
 filter(month==8)

bartlett.test(temp ~ origin,data=weather)
kruskal.test(temp ~ origin,data=weather)
boxplot(temp ~ origin,data=weather)

pairwise.wilcox.test(x = top_5_companies_distances$distance,g = top_5_companies_distances$name)
wilcox.test(humidity_november_LGA$humid,
      mu=50)
```

rm(humidity_november_LGA)

Logistic + CONFUSION Matrix

```
# Goal : predict probability of survival
logistic_train <- glm(Survived ~ .,
            data = train,
            family = "binomial")

summary(logistic_train)
anova(logistic_train)
Odd_ratios <- exp(logistic_train$coefficients)
oddsratio::or_glm(data = train,model = logistic_train)

# Predictions on train set
predicted <- predict.glm(object = logistic_train,
             type = "response")

# Confusion matrix
predicted_train <- data.frame(
 real_class=train$Survived,
 probability=predicted
) %>%
 mutate(predicted_class=case_when(
  probability<0.5 ~ "No",
  probability>=0.5 ~ "Yes"
 ))
          # Confusion Matrix

caret::confusionMatrix(table(predicted_train$real_class,
          predicted_train$predicted_class),positive = "Yes")
```

Plot in a stacked bar plot the repartition of manufactureur by company for the top 5 companies in term of number of flights in 2013.

```
```{r ggplot2, echo=FALSE}

top5_companies <- merge(flights,airlines,by = "carrier") %>%
 group_by(carrier,name) %>%
 summarise(n_flights=n()) %>%
 dplyr::arrange(desc(n_flights)) %>%
 head(5)

count_manufactureur <- planes %>%
 select(tailnum,manufacturer) %>%
 merge(flights[,c("tailnum","carrier")],by="tailnum") %>%
 unique(.) %>%
 merge(airlines,by = "carrier") %>%
 filter(carrier %in% top5_companies$carrier) %>%
 group_by(name,manufacturer) %>%
 summarise(n_planes=n())

ggplot(count_manufactureur,
   aes(fill=manufacturer,
     y=n_planes,
     x=name)) +
 geom_bar(position="fill", stat="identity")+
 theme_bw()
```
```

Anova/Manova

```
fit <- manova(dep_vars ~ Species, data = data)
summary(fit)
anova(fit)
```

MANOVA : Multiple Analysis of Variance

RMSE

```
RMSE[i, 1] <- k[i] # store k-th order
RMSE[i, 2] <- sqrt(sum((fitted(model) - poly_data$y)^2) / length(poly_data$y))
```

RSE

```
RSE <- data.frame(
     order=seq(1:5),
 rse=c(RSE_linear,RSE_quadratic,RSE_cubic,RSE_ordre4,RSE_ordre5)
 )
```

Exponential model

```
model <- lm(y ~ exp(x) + x,
      data=data)

summary(model)

ggplot(data,
     aes(x=x,
       y=y)) +
 geom_point() +
 geom_smooth(formula = y~exp(x)+x, span = 0.75, method = "lm",color="red")
```

```
# Logarithmic model
model <- lm(y ~ log(x) + x,
      data=data)
```

Poisson model + Quasi Poisson + Negative Binomial

```
mod.pois1 <- glm(interlocks~log10(assets),
         family="poisson",
         data=Ornstein)

mod.quasipois1 <- glm(interlocks~log10(assets),
          family="quasipoisson",
          data=Ornstein)

mod.nb1 <- glm.nb(interlocks~log10(assets),
         data=Ornstein)
```

FCA (CA function)

```
fca <- FactoMineR::CA(housetasks)
```

Dimensions (fviz_eig function)

```
fviz_eig(fca)
```

Decision Tree

```
ids <- sample(nrow(Titanic),
        floor(nrow(Titanic)*0.8))

train <- data[ids,]
test <- data[-ids,]

tree <- rpart(survived ~ .,data=train)
```

FAMD

```
res.famd <- FAMD(wine,
         ncp = 5,
         sup.var = NULL,
         ind.sup = NULL,
         graph = TRUE)
```

`Polynomial

```
quadratic_model <- lm(y ~ poly(x, 2), data = poly_data)
```

TESTS

Modeling of a continuous variable :
➢ Linear model (univariate or multivariate)
➢ ANOVA (ANalysis Of Variance), MANOVA, ANCOVA, MANCOVA
➢ Non-linear model (quadratic, cubic, exponential, logarithmic...)
➢ Mixed models
➢ Factorial analysis : PCA

• Modeling of a categorical variable :
➢ Two modalities (Yes vs No) : binary logistic regression
➢ More than two modalities : ordinal or nominal logistic regression
➢ Factorial analysis : CFA, MCA

STATISTICAL TESTS
DEFINITION

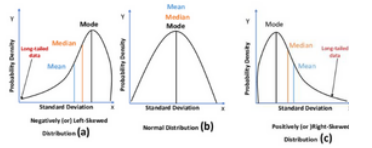
- Statistical **inferential** procedure which allows to test **hypotheses**
- Two tested hypotheses : H_0 (null hypothesis) and H_1 (alternative hypothesis)
- Result : **p-value**, probability to be wrong when rejecting the null hypothesis.
- Two levels of risk : **alpha** (α) and **beta** (β)
- H_0 rejected when pvalue is < α (0.05 or 5%)
- **Two-sided test** (difference) vs **one-sided** (inferiority or superiority)

STATISTICAL TESTS
FAMILIES OF TESTS

- Two families of tests :
  - ➢ **Parametric tests** : require to check assumptions on data (normality of distribution at least)
  - ➢ **Non-parametric tests** : alternative tests when parametric tests are not valid
- Two dimensions of tests :
  - ➢ **Univariate tests** : allows to test hypotheses at the overall level (no groups)
  - ➢ **Multivariate tests** : allows to test hypotheses at group level (2 or more)

- ➢ **Skewness** (asymmetry of distribution)
- ➢ **Kurtosis** (tailedness of distribution)

Normal distribution : **skewness of 0**

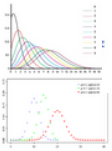


Uniform distribution : runif function (stats package)
Parameters : n = number of random values
min = minimum threshold of values
max = maximum threshold of values

• Normal distribution : rnorm function (stats package)
Parameters : n = number of random values
mean = mean of values
sd = standard-deviation of values

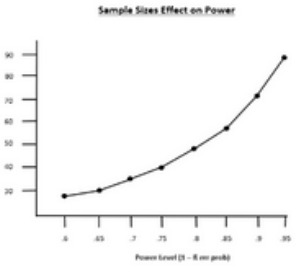
Generation of random variables is easy with R

- Poisson distribution : *rpois* function (*stats* package)
  Parameters : *n* = number of random values
  lambda = shape of the distribution
- Binomial distribution : *rbinom* function (*base* package)
  Parameters : *n* = number of random values
  size = number of trials
  prob = probability of success



STATISTICAL TESTS
STATISTICAL POWER

- Statistical power = 1 - β (beta risk)
- Represents the **ability of a test to detect a significant effect size**
- Strongly correlated with **the number of observations** and **effect size**

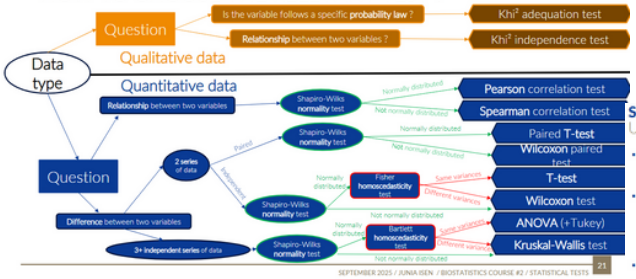


Note: As the sample size increases in the model, so does power.

STATISTICAL TESTS
EXHAUSTIVE LIST OF TESTS

- Quantitative data :
  - ➢ Test the **distribution** of a variable (normality...) : Kolmogorov-Smirnov, Shapiro-Wilks...
  - ➢ Compare the **level** of a variable (mean, median...) **to a reference level**.
  - ➢ Highlight **outliers** (test de Grubbs)
  - ➢ Compare **the level of a variable between two or more groups** (mean, median...) : T-test, univariate ANOVA, Wilcoxon test, Kruskal-Wallis test...
  - ➢ Test the **strength of the relationship between two variables** (correlations)
  - ➢ Compare the **variability of a variable between groups** (variance)
- Qualitative data : Compare occurrence of factors (counts and percentages) between groups (Chi² test)

STATISTICAL TESTS
HOW TO CHOOSE THE RELEVANT TEST ?



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UNIVARIATE TESTS

STATISTICAL TESTS
UNIVARIATE TEST : NORMALITY TEST

- Goal : test if the **distribution of the values** of a quantitative variable **is gaussian** (or normal).
- Hypotheses :
  - ➢ H_0 : the values follow the **normal law**
  - ➢ H_1 : the values do not follow the **normal law**
- In R : function *shapiro.test*(data) (package *stats*)
  if pvalue ≥ 0.05 = data is normally distributed
- Plot for normality : function *ggqqplot*(data) (package *ggpubr*) : if all points are approximately aligned on the diagonal : data is normality distributed

STATISTICAL TESTS
UNIVARIATE TEST : TEST MEAN TO A REFERENCE

- Goal : test if a variable has the same **mean** compared to a **reference value** (**normally distributed variable**)
- Hypotheses :
  - ➢ H_0 : the variable has the same **mean**
  - ➢ H_1 : the variable has a different **mean**
- In R : function *t.test*(*data, mu = reference level*) (package *stats*)
  if pvalue < 0.05 = mean is significantly different from the reference level
- Outputs : test statistic, degrees of freedom, pvalue, 95% confidence interval (CI) of the mean.

STATISTICAL TESTS
UNIVARIATE TEST : TEST MEDIAN TO A REFERENCE

- Goal : test if a variable has the same **median** compared to a reference value (**not-normally distributed variable**)
- Hypotheses :
  - ➢ H_0 : the variable has the same **median**
  - ➢ H_1 : the variable has a different **median**
- In R : function *wilcox.test*(*data, mu = reference level*) (package *stats*)
  if pvalue < 0.05 = median is significantly different from the reference level
- Outputs : test statistic, pvalue

STATISTICAL TESTS
UNIVARIATE TEST : TEST PROPORTION TO A REFERENCE

- Goal : test if the **proportion** of a modality of a binary variable is equal to a target percentage.
- Hypotheses :
  - ➢ H_0 : modality X has the same **proportion**
  - ➢ H_1 : modality X has a different **proportion**
- In R two options :
  - ➢ If small sample (n < 30) : function *binom.test*(x = *number of success*, n = *number of trials*, p = *target proportion*) (package *stats*)
  - ➢ If large sample (n ≥ 30) : function *prop.test*(*same parameters*) (package *stats*)
- Outputs : test statistic, proportion, pvalue, 95% CI of the proportion.

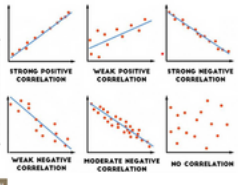
MULTIVARIATE TESTS

CORRELATION

MULTIVARIATE TEST : CORRELATION

- Goal : test if the **coefficient of correlation** between two variables is significantly different from 0.
- Hypotheses :
  - ➢ H_0 : the coefficient is 0
  - ➢ H_1 : the c
- Warning : a coefficient of correlation with a significant pvalue is not necessarily strong ! Correlation > 0.75 in absolute : strong.
- In R : function *cor.test*(x, y, method) (package *stats*)
  Method to use : **pearson** (normality of distributions) vs **spearman**

IATE TEST : CORRELATION



STATISTICAL TESTS
MULTIVARIATE TEST : COMPARISON OF VARIANCES

- Goal : compare the **variance** of a variable between **independents groups** (**homoscedasticity**)
- Hypotheses :
  - ➢ H_0 : the **variances** of each group are equivalent
  - ➢ H_1 : At least one group has a different **variance**
- In R two options (package *stats*) :
  - ➢ If two groups : function *var.test*(x, y, ratio)
  - ➢ If more than two groups : function *bartlett.test*(values ~ group)
- Outputs : test statistic, pvalue

DESCRIPTIVE	INFERENTIAL
It is the analysis of data that helps to describe, show and summarize data under study	It is the analysis of random sample of data taken from a population to describe and make inference about the population
Organize, analyze and present data in a meaningful way	Compares, test and predicts data
It is used to describe a situation	It is used to explain the chance of occurrence of an event
It explain already known data and limited to a sample or population having small size	It attempts to reach the conclusion about the population
Types: Measure of central tendency & Measure of variability	Types: Estimation of parameters & Testing of hypothesis
Results are shown with help of charts, graphs, tables etc.	Results are shown with help of probability scores

STATISTICAL TESTS
MULTIVARIATE TEST : COMPARISON OF MEANS (ANOVA)

- Goal : comparer the **means** of a variable between more **than 2 independent groups** (**normally distributed variable and equivalent variances**)
- Hypotheses :
  - ➢ H_0 : the **means** are equivalent
  - ➢ H_1 : At least two **means** are different
- Need to perform **post-hoc pairwise tests** in case of overall significant pvalue with **Tukey's tests**
- In R function *anova_test*(y ~ group) and *tukey_hsd*(y ~ group) (package *rstatix*)
- Outputs : ANOVA table and pairwise comparisons (for post-hoc tests)

STATISTICAL TESTS
MULTIVARIATE TEST : COMPARISON OF MEANS

- Goal : compare **means** between two **independent groups** (**normally distributed variable and homoscedasticity**)
- Hypotheses :
  - ➢ H_0 : the two **means** are equivalent
  - ➢ H_1 : the two **means** are different
- In R : function *t.test*(x, y, mu) (package *stats*)
- Outputs : test statistic, difference of means, pvalue, 95% CI of the difference

STATISTICAL TESTS
MULTIVARIATE TEST : COMPARISON OF MEANS

- Goal : compare **means** between two **paired groups** (**normally distributed and homoscedasticity**)
- Hypotheses :
  - ➢ H_0 : the two **means** are equivalent
  - ➢ H_1 : the two **means** are different
- Only applicable on **paired** data ("before / after")
- In R function *t.test*(x, y, mu, *paired=T*) (package *stats*)
- Outputs : test statistic, difference of means, pvalue, 95% CI of the difference

UNIVARIATE LINEAR MODELS

$$Y = a \times X + b + \varepsilon$$

Y : variable to explain (continuous)
X : explanatory variable (continuous)
a: coefficient of explanatory variable X(slope of the regression)
b: intercept (value of $Y$when X= 0)
ε : model residuals (proportion of the variability of Ynot explained)

LEAST SQUARES

Adjustment method : least square algorithm: plays on parameters a and b in the equation in order to minimize the sum of squares (sum of ε^2) between real values Y and the regression line (estimated values ŷ)

MULTIVARIATE LINEAR MODELS

MODEL QUALITY

- **Adjustment method** : Least Squares method
- **Quality Model metric** : R^2
- Several R^2 in R :
  - **Classical R^2**
  - **Adjusted R^2** : allows to compare the R^2 of different models with d numbers of parameters
  - **Predictive R^2 (also called Q^2)** : ability of the model to predict new (calculated with PRESS function with cross-validation)

Warning : the more we add parameters the more we increase R^2 (and ov issue)

VARIABLE SELECTION

- Several **methods** of explanatory variables selection :
  - ➢ **Backward** method : from the complete model we iteratively remove the least significant variable (based on the pvalue)
  - ➢ **Forward** method : from the null model (containing only the mean of Y we iteratively test each variable add the most significant one (based on the pvalue)
  - ➢ **Stepwise** method : mix of the two other methods

The algorithm stops when it can not find another variable to include in the model (based of a pvalue threshold : 15% or 0.15 by default)

In R : *stepAIC* function included in package *MASS*

CROSS VALIDATION

- Statistical procedure often used in Machine Learning which allows to reduce the **overfitting issue**, very harmful during the development of **predictive models**.
- Idea : a model is **trained** on a representative subset of data and **tested** on the remaining data (not seen) several times.
- **Predictive R^2** is calculated with cross-validation : the model is built on all data except one and tested on this datapoint. This step is done N times (N=number of observations). This method is called LOOCV (Leave-One Out Cross Validation)
- In R many techniques available in *caret* package

MANOVA (MULTIVARIATE ANOVA)

HIGHLIGHTS

Goal : assess impact of one (or many) categorical parameter (factor) on two (or more) continuous outcomes

• Assumptions :
➢ dependent variables should be normally distributed within groups (checked with mshapiro.test function in mvnormtest package)
➢ Homogeneity of variances across the range of predictors
➢ No multicollinearity between dependent variables : checked with cor_test function (package rstatix)

ANCOVA (ANALYSIS OF COVARIANCE)

Similar to ANOVA but with covariate(s) (continuous factor)
• Assumptions :
➢ The covariate(s) and the factor are independent
➢ Homogeneity of variances across the modalities of the factor
➢ Homogeneity of regression slopes within each group
• Need to perform post-hoc pairwise tests in case of overall significant pvalue with Tukey's tests
• In : function manova(y ~ factor + covariate) and tukey_hsd(y ~ group) (package rstatix)

COEFFICIENT OF DETERMINATION (R^2)

- **Quality model metric** : Coefficient of Determination R^2

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

with y_i: real value for observation i
$\hat{y}_i$: predicted value by the model for observation i
$\bar{y}$: mean of real values

- R^2 = square of **linear coefficient of correlation**
- Allows to determine the proportion of variability of Y explained by the model (in %)
- **Warning** : a good R^2 does not mean the model has a good **predictive accuracy**!

MULTICOLINEARITY

- Statistical issue where two (or more) explanatory variables are **strongly correlated**
- Including such variables in a model can rise **instability of parameters** of the equation and also cause a **non-convergence of variable selection algorithms**
- This high variability of parameters **does not impact predictive ability** of the model by can **strongly impact significance of factors** (pvalues).
- The **VIF** index (Variance Influence Factor) is useful for highlighting variables which are highly correlated. VIF is available in R with the ***car*** package with ***vif*** function.
- Remove a variable when the VIF is > 5.

NON-LINEAR MODELS

Central limit theorem (CLT) : under appropriate conditions the distribution of a normalized version of the sample mean converges to a standard normal distribution.

QUADRATIC MODEL :
Adjustment method : least square algorithm: plays on parameters a, b and c in the equation in order to minimize the sum of squares (sum of ε^2) between real values Y and the model (estimated values ŷ)

ANOVA, MANOVA, ANCOVA AND MANCOVA

ONE-WAY ANOVA
COMPARISON OF MEANS BETWEEN GROUPS (ANOVA)

- Goal : compare **means** of a variable between more **than 2 independent groups** (normally distributed variable and equivalent variances)
- Hypotheses :
  - ➢ H_0 : the **means** are equivalent
  - ➢ H_1 : At least two **means** are different
- Need to perform **post-hoc pairwise tests** in case of overall significant pvalue with **Tukey's tests**
- In R function *aov*(y ~ group) and *tukey_hsd*(y ~ group) (package *rtatix*)
- Outputs : ANOVA table and pairwise comparisons (for post-hoc tests)

TWO-WAY ANOVA
COMPARISON OF MEANS ACROSS SUBGROUPS

- Goal : assess impact of two (or more) categorical parameters (factor) on a continuous outcome (normally distributed variable and equivalent variances)
- Many hypotheses (2 factors) :
  - ➢ H_{A0} : no difference in the means of factor A
  - ➢ H_{A1} : means are not equal for factor A
  - ➢ H_{B0} : no difference in the means of factor B
  - ➢ H_{B1} : means are not equal for factor B
  - ➢ H_{AB0} : no interaction between factors A and B
  - ➢ H_{AB1} : significant interaction between factors A and B
- In R function *anova_test*(y ~ factors) and *tukey_hsd*(y ~ parameters) (package *rtatix*)

GENERALIZED LINEAR MODELS

When equations are too complex and not linear, Least Square method is not the most converging algorithm to use when trying to fit a model to data.

• A powerful alternative way exists : Maximum Likelihood Estimation (MLE)

• Principle : the algorithm try to find the parameters of the distribution of data (and not trying to minimize residuals)

GENERALIZED LINEAR MODELS
INTRODUCTION

Two main models using MLE, also called "**Generalized Linear Models**" exists and aim to link a discrete Y to continuous and/or discrete factors :

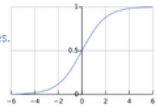
- **Poisson model** (and alternative models : quasi-poisson, negative-binomial) : often use to model **count data**
- **Logistic regression** : commonly used in the research of **risk factors of a disease** in clinical trials for example

These two models are available in R in with ***glm*** function (package ***stats***).

LOGISTIC REGRESSION (binary variable)

LOGISTIC REGRESSION

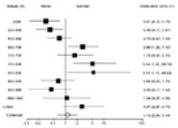
- **Goal** : explain a binary variable Y with a continuous or discrete variable(s) X
- **Assumptions** : no colinearity between explanatory variables, independance of observations, no outliers, binary response well balanced between the two classes.
- **Link function** : logit
- No residuals normality study : **predicted vs observed classes table**
- Results : $\hat{Y}$: **probability** to belong to group 1
  Odds-ratios : **effect size** of each explanatory variable



LOGISTIC REGRESSION – ODDS-RATIOS (OR)

$$\text{Odds} - \text{ratio} = \frac{\frac{p}{1-p}}{\frac{q}{1-q}} = \frac{p(1-q)}{q(1-p)}$$

with : p : probability of event in group A
q : probability of event in group B



- Values (always positive) : < 1 : event is less frequent in group A
  = 1 : no difference in frequency of event in both groups
  > 1 : event is more frequent in group A
- OR is always presented with **95% confidence interval** (95%CI) and pvalue

OTHER FACTOR ANALYZES (FCA, MCA & FAMD)

FACTOR CORRESPONDENCE ANALYSIS (FCA) : study the link (kind of correlation) between two categorical variables

FACTOR CORRESPONDENCE ANALYSIS (FCA)
INPUT DATA

Three ways to code data :

- **Contingency tables** : count data
- **Complete disjunctive table** : binary variables (0 vs 1 for each modality)
- **Burt's table** : sum of counts from a complete disjunctive table (symmetric matrix)

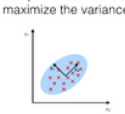


MULTIPLE CORRESPONDENCE ANALYSIS (MCA) : study the link between several categorical variables (extension of FCA)

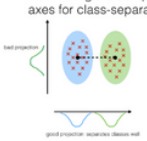
FACTOR ANALYSIS OF MIXED DATA (FAMD): study the link between categorical variables and continuous variables

LINEAR DISCRIMINANT ANALYSIS (LDA)
INTRODUCTION

PCA: component axes that maximize the variance



LDA: maximizing the component axes for class-separation



find the best linear combination of factors which draws the most discriminant boundaries between groups.

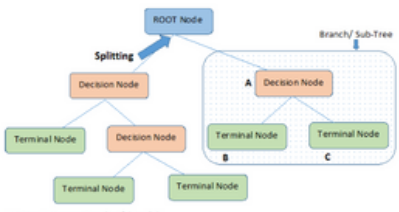
DECISION TREES
PRUNING TREES

Why don't use the deepest tree ?

1. Prevents **Overfitting**
2. Improves **Generalization**
3. Reduces Model **Complexity**
4. Enhances **Interpretability**
5. Speeds Up **Training** and **Inference**
6. Facilitates Model **Maintenance**



DECISION TREES



Note:- A is parent node of B and C.

Thanks to Maximum Likelihood function, **two indexes** are available for model comparisons :

➢ **Akaike information criterion** (AIC) : trade-off between the goodness of fit of the model and the simplicity of the model: balance between overfitting and underfitting.

$$AIC = 2 \times k - 2 \times \ln(\hat{L})$$
with k : number of parameters (X)
$\hat{L}$: Maximum Likelihood value

Lower AIC : best model

➢ **Bayesian information criterion** (BIC) : like AIC but with penalization of complex models when the number of points is > 8.

$$BIC = k \times \ln(n) - 2 \times \ln(\hat{L})$$
with k : number of parameters (X)
n : number of observations
$\hat{L}$: Maximum Likelihood value

Lower BIC : best model

Output :
Coefficients of the equation
Residuals
Fitted values
Dispersion analysis
AIC

LOGISTIC REGRESSION – QUALITY OF MODEL

- No R^2 in logistic regression
- Instead, a **confusion matrix** is used :

Predicted / Actual	Sick	Healthy
Positive test	TRUE positive (TP)	FALSE positive (FP)
Negative test	FALSE negative (FN)	TRUE negative (TN)

Specificity $= \frac{TP}{TP+FN}$
(True Positive Rate : probability the test is positive in the **sick** pop)

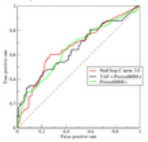
Sensitivity $= \frac{TN}{TN+FP}$
(False Positive Rate : probability the test is **negative** in the **healthy** pop)

Positive Predicted Value $= \frac{TP}{TP+FP}$
(probability the illness is present when test is positive)

Negative *Predicted Value* $= \frac{TN}{TN+FN}$
(probability the illness is absent when test is negative)

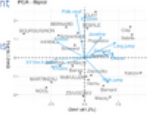
LOGISTIC REGRESSION – ROC CURVE

- Helpful tool to **compare models** : **ROC curve** (Receiver Operative Characteristic)
- Displays the **evolution of TPR and FPR** in the population
- Dataset is sorted by ascending **score** (probability) calculated by the logistic regression and **TPR** and **FPR** are calculated
- The **Area Under the Curve** (AUC) is calculated (from 0 to 1) : **roc** from pROC package
- The diagonal represents the **random classification** of individuals (AUC of 0.5).



PRINCIPAL COMPONENT ANALYSIS (PCA)
PLOTS : BIPLOT

- Allows to visualize the simultaneously **coordinates of each individual and variables** on the most important factorial planes.
- The coordinates are **proportional** to the contributions of each factor on each axis.
- **Opposite directions** mean significant different results for individuals.
- In this example, there is no **clearly visible clusters** of individuals based on their results.

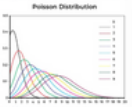


Classification	Clustering
Uses labelled data as the input	Uses unlabelled data as the input
The output is known	The output is unknown
Uses supervised machine learning	Uses unsupervised machine learning
A training data set is provided and used to produce classifications	A training data set is not provided and used to produce clusters
Examples of algorithms: Decision-trees, Bayesian Classifiers and Support Vector Machines (SVM)	Examples of algorithms: Partition-based clustering (k-means), Hierarchical clustering (agglomerative & divisive) and DBSCAN
Can be more compex than clustering	Can be less compex than classification
Does not specify areas for improvement	Specifies areas for improvement
Two-phase	Single-phase
Boundary conditions must be specified	Boundary conditions do not always need to be specified

k-Means Clustering	k-Nearest Neighbor (k-NN)
Unsupervised learning	Supervised learning
Grouping similar data points into clusters	Classifying or predicting based on nearest neighbors
No labeled data required	Requires labeled training data
Iterative process	Computationally intensive at prediction time
Centroids and cluster assignments	Predicted labels or values

POISSON MODEL (discrete variable) glm

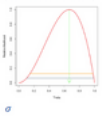
- **Goal** : explain a discrete variable Y (counts) with a variable X
- **Hypothesis to test** : Y follows a poisson distribution (with parameter λ = mean and standard-deviation)
- **Equation** : $Y = a \times X + b + \varepsilon$



with Y : variable to explain (discrete)
X : explanatory variable (continuous ou discrete)
a : coefficient of X
b : intercept (value of Y when $X = 0$)
ε : model residuals (proportion of the variability of Y not explained)

- **Adjustment method** : **Maximum Likelihood Algorithm** : the model will try to guess the value of the parameter $\lambda = \exp(X \times \beta)$ in order to minimize the sum of squares (sum of ε^2) between real values Y and the model (estimated values $\hat{Y}$)
- **Equation** : $P(Y=y|\lambda) = \frac{e^{-\lambda}\lambda^y}{y!}$

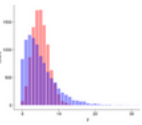
with Y : variable to explain (discrete)
y : observations (continuous ou discrete)
λ : parameter of Poisson distribution



Validity : distribution of Y values follows a Poisson law : $\bar{y} = \sigma$

GENERALIZED LINEAR MODELS
POISSON MODEL

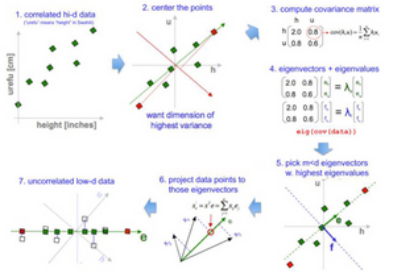
- **Overdispersion** : the observation that variation is higher than would be expected according to the Poisson law (assessed when $\frac{\text{residual deviance}}{ddl} > 1$)



- In case of overdispersion : **quasi-poisson** or **negative-binomial** model
- R^2 of Generalized Linear Models does not have the same meaning compared to linear model : the % represents the **% of reduction of deviance** (difference between log-likelihood functions) of the current model vs the **null model** (with only the intercept in the equation).

QUASI-POISSON MODEL glm
NEGATIVE-BINOMIAL MODEL glm.nb

PRINCIPAL COMPONENT ANALYSIS (PCA): powerful reduction-dimension statistical tool



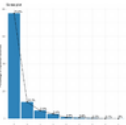
PRINCIPAL COMPONENT ANALYSIS (PCA)
PRINCIPAL COMPONENTS

- Principal components (PC) are **linear combinations** of variables.
- Need to **scale values** : variables with **high variance** and/or high **values** can be too influent.
- Each PC has its own **eigenvalue** calculated on the covariance matrix
- K variables = K axes and number of spaces $= \frac{K!}{2 \times (K-2)!}$
- PCA not useful when **non-linear relationship** between parameters is noticed



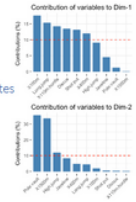
PRINCIPAL COMPONENT ANALYSIS (PCA)
PLOTS : SCREE PLOT

- Allows to visualize **eigenvalues of PCs**
- Useful for the choice of the **number of PCs** to keep in the analysis
- In this example, the first factorial plane (PC1 + PC2) explains **almost 90% of the information contained in the dataset**



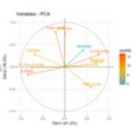
PRINCIPAL COMPONENT ANALYSIS (PCA)
PLOTS : VARIABLES CONTRIBUTION

- Allows to visualize the **most influent variables** for the building of each PC
- The **contributions are proportional** to the coordinates of each factor on the studied axis.
- In this example, **the second factorial plane** (PC2) **is highly impacted by the two first variables** which explains more than 60% of the variability.



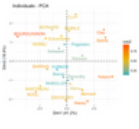
PRINCIPAL COMPONENT ANALYSIS (PCA)
PLOTS : VARIABLES PLOT

- Allows to visualize the **coordinates of each variable** on the most important factorial planes.
- The coordinates are **proportional** to the contributions of each factor on each axis.
- **Opposite directions** mean significant different results for individuals.
- In this example, the two extreme variables on PC1, sprint trials (100m, 100h hurdle) results are opposite to long and high jump, **meaning individuals with high performances in sprint usually don't success in jumping trials.**

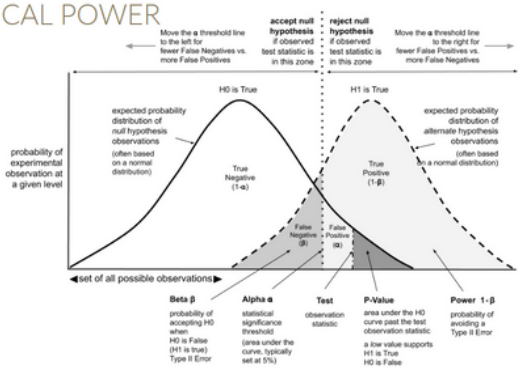


PRINCIPAL COMPONENT ANALYSIS (PCA)
PLOTS : INDIVIDUALS PLOT

- Allows to visualize the **coordinates of each individual** on the most important factorial planes.
- The coordinates are **proportional** to the contributions of each factor on each axis.
- **Opposite directions** mean significant different results for individuals.
- In this example, there is no **clearly visible clusters** of individuals based on their results.

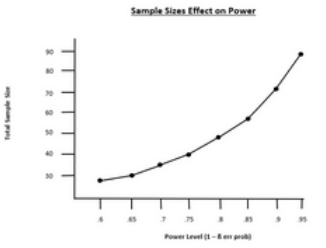


SAMPLE SIZE CALCULATION
STATISTICAL POWER



SAMPLE SIZE CALCULATION
STATISTICAL POWER

- Statistical power = 1 - β (beta risk)
- Represents the ability of a test to detect a significant effect size
- Strongly linked with:
  - Sample size
  - Effect size
  - Significance level



Note: As the sample size increases in the model, so does power.

SAMPLE SIZE CALCULATION
APPLICATION WITH R

Package **pwr** contains **many functions for power / sample size calculation** :

- **pwr.2p.test :** test of **two proportions** (same N in groups)
- **pwr.2p2n.test :** test of **two proportions** (unequal N in groups)
- **pwr.p.test** : Proportion (one sample)
- **pwr.r.test** : Correlation
- **pwr.chisq.test** : Chi-Square test
- **pwr.f2.test** : General Linear Model
- **pwr.anova.test** : balanced one-way ANOVA
- **pwr.norm.test** : Test of a mean to a reference (known variance)
- **pwr.t.test** : T-tests (one sample, two samples, paired)
- **pwr.t2n.test** : T-tests (two samples with unequal N)

Same way to use: all parameter values provided **except one** (result of the function)

pwr.2p.test function (*pwr* package)
Parameters : h = effect size : $2 \times \arcsin(\sqrt{p_1}) - 2 \times \arcsin(\sqrt{p_2})$
n = common sample size
sig.level = significance level (default : 5%)
power = power (in %)

pwr.2p2n.test function (*pwr* package)
Parameters : h = effect size : $2 \times \arcsin(\sqrt{p_1}) - 2 \times \arcsin(\sqrt{p_2})$
n1 = sample size group 1
n2 = sample size group 2
sig.level = significance level (default : 5%)
power = power (in %)

pwr.chisq.test function (*pwr* package)
Parameters : w = effect size : $\sqrt{\sum_{i=1}^{m} \frac{(p0_i - p1_i)^2}{p0_i}}$
with m : number of cells
$p0_i$: cell probability in ith cell under H0
$p1_i$: cell probability in ith cell under H1
N = Total sample size
df= degrees of freedom (number of categories -1)
sig.level = significance level (default : 5%)
power = power (in %)

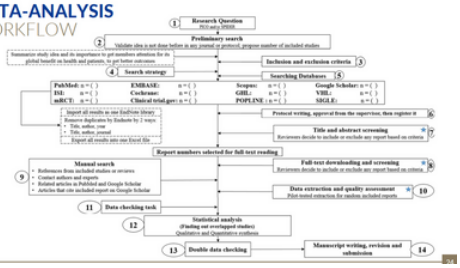
Package *effectsize* useful for extracting f effect size from an **test** (function *cohens_w*)

pwr.p.test function (*pwr* package)
Parameters : h = effect size : $2 \times \arcsin(\sqrt{p_1}) - 2 \times \arcsin(\sqrt{p_2})$
n = sample size
sig.level = significance level (default : 5%)
power = power (in %)

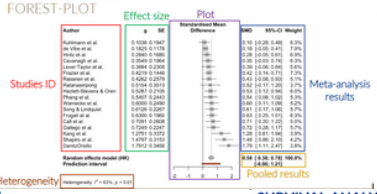
pwr.r.test function (*pwr* package)
Parameters : *n* = Total sample size
r = correlation
sig.level = significance level (default : 5%)
power = power (in %)

META-ANALYSIS
BIAS :
1. Information bias : key study variables are inaccurately measured or classified
2. Interviewer bias : way an interviewer ask questions or react to responses
3. Publication bias : decision to publish research findings is based on their nature or the direction of their results
4. Research bias : researcher's beliefs or expectations influence the research design or data collection process
5. Response bias : respondents tend to provide inaccurate or false answers to self-report questions
6. Selection bias : sampling bias, attrition bias, survivorship bias...
7. Other bias : cognitive bias, confirmation bias, framing effect, halo effect, placebo effect...

META-ANALYSIS
WORKFLOW



META-ANALYSIS
FOREST-PLOT



META-ANALYSIS
HETEROGENEITY

- The more we include studies in a meta-analysis, the more we increase the **variability of measures** (bias...) and then increase **heterogeneity : clinical** diversity, **methodological** diversity, **statistical** heterogeneity.
- How to evaluate **heterogeneity** in a meta-analysis :

I^2 statistic : $I^2 = \left(\frac{Q-df}{Q}\right) \times 100\%$ with Q : Chi^2 statistic and df = degrees of freedom

Value	Interpretation
0% to 30%	Low heterogeneity
30% to 50%	Moderate heterogeneity
50% to 75%	Substantial heterogeneity
	Considerable heterogeneity

LONGITUDINAL DATA ANALYSIS
LONGITUDINAL DATA OVERVIEW

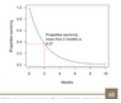
- **Longitudinal data** : collection of data for an individual across time.
- During **clinical trials** (except retrospective trials), data is collected, monitored and stored in **secured database** for each patient included in the trial.
- This data can have **various forms** : **biomarkers** (in blood, urine, stood...), **electronic health records** (heart rate, respiratory rate, temperature...), quality of life **questionnaires**...
- Each patient is anonymized with **a unique Patient ID**.
- Data collection, treatment and storage is **strictly controlled during a clinical trial**.

SURVIVAL ANALYSIS
KAPLAN-MEIER ESTIMATOR

- **Definition** : Non-parametric statistic used to estimate the **survival function** from lifetime data.
- Invented by **Edward L. KAPLAN** (1920 – 2006) and **Paul MEIER** (1924 – 2011), two American statisticians



- Survival function : $S(t) = P(T > t)$



- **Kaplan-Meier plot** : illustrates the evolution of **overall** survival or within groups across time.
- **Log-rank test** : compare the **survival of two groups** with a pvalue (also called Cochran-Mantel-Haenszel test).

SURVIVAL ANALYSIS
KAPLAN-MEIER PLOT

